



## Stagnone di Marsala digital twin: project resumption and status

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### 1. Summary

Work on the coupled three-dimensional model resumed on 5 August after a pause since mid-June, due to the rewriting of the manuscripts related to GEE. As we have several runs already done, the main outcome of the past three days is that the first manuscript now has a complete draft, in LaTeX, formatted for *Estuarine, Coastal and Shelf Science* and within all of that journal's limits. A numerical problem that had cost us two of the eight experiments was also traced and fixed today, and the corresponding re-runs are in progress (Section 3).

The manuscript is provisionally titled “**Transport validation of a wave-coupled lagoon model: interacting effects of seagrass roughness and bed mobility**”, with the three of us as authors. I would be grateful for your comments on both the framing and the content.

### 2. What the study asks and what it found

Considering that coastal lagoon models are routinely validated against water level and then used to describe transport, I set the paper to ask whether the first licenses the second. Six model configurations were run over the same nine days of July 2025, during the drifters deployments, from the same initial condition and forcing, differing only in wave coupling, bed mobility, and whether seagrass resistance is uniform or distributed from a satellite classification. Each was scored against three tide gauges and against 35 GPS surface drifters released over twelve deployments.

**Water level does not separate the configurations.** In the results, WL anomaly RMSE differs by six millimetres or less between the runs at any station, which is within the measurement uncertainty, and no configuration is consistently best across the three stations. Over the same simulations, using Liu–Weisberg skill score, we obtained values spanning from 0.505 to 0.655 and endpoint separation spans 334 to 661 m. A calibration conducted on water level alone would have had no basis on which to prefer any of them.

**No treatment improves Lagrangian skill on its own.** Distributed seagrass roughness on a fixed bed changes skill by 0.009 or less, bed mobility under uniform roughness by 0.019, and wave coupling reduces it by 0.035 to 0.042. Applied together, distributed roughness and bed mobility raise skill by 0.131 to 0.149, improving 32 of the 35 drifters, and only that combination reproduces the observed transport rate.

**Splitting the trajectory error into magnitude and direction shows what each treatment corrects**, which is the most recent result and the one I consider the paper's contribution.

| Configuration                     | Path length<br>(sim/obs) | Net displacement<br>(sim/obs) | Heading error<br>(deg) |
|-----------------------------------|--------------------------|-------------------------------|------------------------|
| Four fixed-bed members            | 0.77–0.80                | 0.78–0.82                     | 15.8–16.2              |
| Mobile bed, uniform roughness     | 0.94                     | 0.95                          | 12.5                   |
| Mobile bed, distributed roughness | 0.98                     | 0.99                          | <b>8.4</b>             |

The four fixed-bed members coincide in all three measures. Bed mobility acts on magnitude, restoring the distance travelled. Distributed roughness acts on direction, and does so only once the bed can move: on a fixed bed it leaves the heading error unchanged to the first decimal. This is reported as a narrowing rather than a complete mechanism, since reconstructing endpoint separation from a mean

displacement ratio and a mean heading still underpredicts the two mobile-bed members by a factor of 1.4 to 1.9.

Until now, I could not find a simpler explanation: the spatial pattern of bed change is indistinguishable between the two mobile-bed members, canopy drag does not suppress erosion beneath the meadows, and the successful configuration does not carry a stronger mean current field.

Two further results support the modelling choices. The canopy parametrisation reduces mean wave orbital velocity at the bed by 14.3% over *Posidonia*, 13.7% over *Cymodocea*, 11.3% over sand and 3.3% over exposed rock, an ordering that follows canopy density. And surface speed exceeds bed speed by a factor of 1.9 to 4.5 at sub-metre mean depth, consistent with De Marchis et al. (2012) here and the quantitative case for solving in three dimensions.

### 3. An item that closed while this was being written

The design is a factorial in three factors, and until today only six of its eight cells had been obtained. The two missing cells combine a mobile bed with no wave coupling, and both aborted on the model's velocity cap, at 26.96 and 43.42  $\text{ms}^{-1}$  against a limit of 25. They did so deterministically: two independent attempts stopped at the same simulated instant with velocities identical to six decimal places. The manuscript as it stands reports this as a property of the system, under a heading stating that a mobile bed does not stabilise without waves.

That reading was wrong, and I am glad the check was made before submission. The instability turned out to be offshore and diffuse rather than a defective cell: of the 150 fastest cells before the abort, none was inside the lagoon, none was above the datum, and their median depth was 34 m.

The cause is the coupling between suspended sediment and density. The morphology settings had suspended sediment feeding the density field, and therefore the baroclinic pressure gradient, which is the only route by which sediment can move momentum in a deep cell holding no erodible material. It also explains an earlier fix that failed: confining erodible thickness to depths above 20 m changed nothing, because that bounds erosion and not the suspended load arriving by advection. With the coupling disabled the run passed the abort instant without incident and is now at 65% of the nine days, with a time step almost double what it had before.

Three re-runs are under way so that all four mobile-bed members share the setting, and should finish overnight. The design then closes at eight cells, and the effect of wave coupling becomes measurable on a mobile bed, which the manuscript currently states cannot be done. Sections 3.6, 4.3, 5 and 6 will be revised, and the subsection asserting that a mobile bed does not stabilise without waves will be removed rather than reworded. I do not expect the results in Section 2 to move, since they rest on members that were never affected, but they will be recomputed and checked.

### 4. Manuscript status and what I would ask of you

The draft is complete across all six sections, with eight figures and three tables, 6200 words of body text against a limit of 8000, and 24 references verified against the source PDFs. Highlights and a cover letter are drafted. Outstanding are a graphical abstract, which *ECSS* requires.

I would welcome your reading on three points in particular:

1. Whether the framing holds. The paper claims a validation-strategy result, trying not to sit just as an study case of the Stagnone, since the journal does not accept manuscripts of mainly local interest. Did you agree?
2. Whether reporting the bed-mobility result is wise given that its critical shear stress for erosion has



not been constrained against a repeated bathymetric survey. We are using the old bathymetry we have, maybe the results are better for mobile beds because it changes the morphology near the nowadays morphology, and if we try again with updated data, the results could be completely different. The manuscript states this limitation openly, and the lagoon bed changes by up to 1.0 m over nine days in isolated cells, which is not a plausible rate. There's a chance that we obtain the new bathymetry soon, and if so, we should wait to rerun the simulations with it before submission.

3. The CRediT roles I have assigned, which follow those of the reservoirs manuscripts.

The full draft, figures and analysis code are in the project repository, and I can send the compiled PDF on request.

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